

Laboratory Work #2 — Normalization

Task 1

The table `Booking_receipt`(`booking_id`, `passenger_full_name`, `passenger_passport_number`, `flight_number`, `departure_airport_name`, `departure_city`, `arrival_airport_name`, `arrival_city`, `airline_name`, `seat_numbers`, `ticket_price`) violates 1NF, 2NF and 3NF at the same time. Each violation is explained below.

1NF

The column `seat_numbers` stores a comma-separated list of values (e.g. "12A, 12B") whenever a booking covers several passengers on the same reservation. First Normal Form requires that every attribute hold a single, atomic value — a cell cannot contain a list. Because `seat_numbers` can contain several seat codes at once, the table is not in 1NF.

2NF

After splitting `seat_numbers` so that each row represents one seat (see Task 2), the natural primary key becomes the composite key (`booking_id`, `seat_number`). Attributes such as `passenger_full_name`, `passenger_passport_number`, `flight_number`, `departure_airport_name`, `departure_city`, `arrival_airport_name`, `arrival_city`, `airline_name` and `ticket_price` depend only on `booking_id` — they do not depend on `seat_number` at all. This is a partial functional dependency (dependency on part of the composite key, not the whole key), which is exactly what 2NF forbids.

3NF

Even after removing the partial dependencies, a transitive dependency remains inside the resulting Booking-level table: `flight_number` determines `airline_name`, `departure_airport_name`, `departure_city`, `arrival_airport_name` and `arrival_city`, while `booking_id` determines `flight_number`. So the chain `booking_id` → `flight_number` → `airline_name` (and the other flight-related columns) is transitive: those columns depend on `flight_number`, a non-key attribute, rather than directly on the primary key `booking_id`. This violates 3NF.

Task 2

UNF

Booking_receipt (unnormalized)

Attribute	Key	Notes
<code>booking_id</code>		repeats for multi-passenger/multi-seat bookings
<code>passenger_full_name</code>		
<code>passenger_passport_number</code>		
<code>flight_number</code>		
<code>departure_airport_name</code>		
<code>departure_city</code>		
<code>arrival_airport_name</code>		
<code>arrival_city</code>		
<code>airline_name</code>		
<code>seat_numbers</code>		comma-separated list, e.g. "12A, 12B" — violates atomicity
<code>ticket_price</code>		

1NF

Action: remove the repeating group by giving each seat its own row. The comma-separated list in seat_numbers is split into individual seat_number values, duplicating the other attributes for every seat that belongs to the same booking.

Booking_receipt_1NF

Attribute	Key	Notes
<u>booking_id, seat_number</u>	PK	composite primary key
passenger_full_name		
passenger_passport_number		
flight_number		
departure_airport_name		
departure_city		
arrival_airport_name		
arrival_city		
airline_name		
ticket_price		

2NF

Action: remove partial dependencies. Every attribute that depends only on booking_id (not on the full composite key booking_id + seat_number) is moved into its own table, leaving only the seat assignment itself keyed on the full composite key.

Booking (2NF)

Attribute	Key	Notes
<u>booking_id</u>	PK	
passenger_full_name		
passenger_passport_number		
flight_number	FK	
airline_name		
departure_airport_name		
departure_city		
arrival_airport_name		
arrival_city		
ticket_price		

Seat_assignment (2NF)

Attribute	Key	Notes
<u>booking_id, seat_number</u>	PK	composite primary key; booking_id is also FK

3NF

Action: remove transitive dependencies. In Booking (2NF), airline_name, departure_airport_name, departure_city, arrival_airport_name and arrival_city all depend on flight_number rather than directly on booking_id. These columns are moved into a new Flight table keyed on flight_number, and Booking keeps only flight_number as a foreign key.

Booking (3NF)

Attribute	Key	Notes
<u>booking_id</u>	PK	
passenger_full_name		
passenger_passport_number		
flight_number	FK	references Flight(flight_number)
ticket_price		

Flight (3NF)

Attribute	Key	Notes
<u>flight_number</u>	PK	
airline_name		
departure_airport_name		
departure_city		
arrival_airport_name		
arrival_city		

Seat_assignment (3NF, unchanged from Step 2)

Attribute	Key	Notes
<u>booking_id, seat_number</u>	PK	booking_id is FK → Booking(booking_id)

Task 3

The following functional dependencies justify each normalization step above:

- booking_id → passenger_full_name, passenger_passport_number, flight_number, ticket_price
- flight_number → airline_name
- flight_number → departure_airport_name, departure_city, arrival_airport_name, arrival_city
- (booking_id, seat_number) → (no further non-key attribute; this composite key only identifies the seat assignment row itself)

Booking_id → flight_number together with flight_number → airline_name forms the transitive chain booking_id → flight_number → airline_name that was removed in the 3NF step (and likewise for the other flight/airport/city columns).

Task 4

Current primary key: flight_id.

Functional dependencies in Flights:

- flight_id → sch_departure_time, sch_arrival_time, departing_airport_id, arriving_airport_id, departing_gate, arriving_gate, airline_id, act_departure_time, act_arrival_time, created_at, updated_at

Transitivity check: departing_airport_id, arriving_airport_id and airline_id are foreign keys that reference the Airport and Airline tables; they do not determine any other attribute that remains stored inside Flights itself (airport names, cities and airline names live only in the Airport and Airline tables, not duplicated in Flights). Because none of the non-key attributes of Flights depend on another non-key attribute of Flights, there is no transitive dependency, so Flights is already in 3NF.

Task 5

The SQL below implements the 3NF design from Task 2 and populates it with 5 rows split out from a small denormalized sample dataset.

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Query

```
1 SELECT
2   tc.table_name,
3   kcu.column_name AS primary_key_column
4 FROM information_schema.table_constraints tc
5 JOIN information_schema.key_column_usage kcu
6   ON tc.constraint_name = kcu.constraint_name
7 WHERE tc.constraint_type = 'PRIMARY KEY'
8   AND tc.table_schema = 'public'
```

Data Output

table_name	primary_key_column_name
airline	airline_id
airport	airport_id
baggage	baggage_id
baggage_ch...	baggage_check_id
boarding_pa...	boarding_pass_id
booking	booking_id
booking_flg...	booking_flight_id
booking_nor...	booking_id
flight	flight_number
flight_norm	flight_number
flights	flight_id
passengers	passenger_id
seat_assign...	seat_number
seat_assign...	booking_id
seat_assign...	booking_id
seat_assign...	seat_number
security_che...	security_check_id

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Successfully run. Total query runtime: 51 msec. 17 rows affected.

Total rows: 17 Query complete 00:00:00.051

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Query

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1 SELECT
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4 FROM information_schema.table_constraints tc
5 JOIN information_schema.key_column_usage kcu
6   ON tc.constraint_name = kcu.constraint_name
7 WHERE tc.constraint_type = 'PRIMARY KEY'
8   AND tc.table_schema = 'public'
```

Data Output

table_name	primary_key_column_name
airline	airline_id
airport	airport_id
baggage	baggage_id
baggage_ch...	baggage_check_id
boarding_pa...	boarding_pass_id
booking	booking_id
booking_flg...	booking_flight_id
booking_nor...	booking_id
flight	flight_number
flight_norm	flight_number
flights	flight_id
passengers	passenger_id
seat_assign...	seat_number
seat_assign...	booking_id
seat_assign...	booking_id
seat_assign...	seat_number
security_che...	security_check_id

Showing rows: 1 to 17 Page No: 1 of 1

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Query

```
SELECT conname, conrelid::regclass AS table_name, conrelid::regclass AS references_table
FROM pg_constraint
WHERE contype = 'f'
ORDER BY table_name;
```

Data Output

conname	table_name	references_table
name	regclass	regclass
fk_baggagecheck_passenger	baggage_ch...	passengers
fk_baggagecheck_booking	baggage_ch...	booking
fk_baggage_booking	baggage	booking
fk_bookingflight_booking	booking_flg...	booking
fk_bookingflight_flight	booking_flg...	flights
fk_booking_passenger	booking	passengers
fk_security_passenger	security_che...	passengers
fk_boardingpass_booking	boarding_pa...	booking
seat_assignment_booking_id_fkey	seat_assign...	booking
booking_norm_flight_number_fkey	booking_nor...	flight_norm
seat_assignment_norm_booking_id_f...	seat_assign...	booking_norm
fk_flights_departing_airport	flights	airport
fk_flights_arriving_airport	flights	airport
fk_flights_airline	flights	airline

Showing rows: 1 to 14 Page No: 1 of 1

Successfully run. Total query runtime: 55 msec. 14 rows affected.

Total rows: 14 Query complete 00:00:00.055

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Query

```
SELECT table_name
FROM information_schema.tables
WHERE table_schema = 'public'
ORDER BY table_name;
```

Data Output

table_name
name
airline
airport
baggage
baggage_ch...
boarding_pa...
booking
booking_flg...
booking_nor...
flight
flight_norm
flights
passengers
seat_assign...
seat_assign...
security_che...

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Successfully run. Total query runtime: 48 msec. 15 rows affected.

Total rows: 15 Query complete 00:00:00.048

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airport

baggage

baggage_check

boarding_pass

booking

booking_flight

booking_norm

flight

flight_norm

flights

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No limit

Query

Query History

Scratch Pad

1

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3

4

SELECT table_name, column_name, is_nullable

FROM information_schema.columns

WHERE table_schema = 'public'

ORDER BY table_name, ordinal_position;

Data Output

Messages

Notifications

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	table_name	column_name	is_nullable
	name	name	character varying (3)
38	booking	ticket_price	NO
39	booking_flight	booking_flight_id	NO
40	booking_flight	booking_id	NO
41	booking_flight	flight_id	NO
42	booking_flight	created_at	NO
43	booking_flight	updated_at	NO
44	booking_norm	booking_id	NO
45	booking_norm	passenger_full_name	NO
46	booking_norm	passenger_passport_num	NO
47	booking_norm	flight_number	NO
48	booking_norm	ticket_price	NO
49	flight	flight_number	NO
50	flight	airline_name	NO
51	flight	departure_airport_name	NO
52	flight	departure_city	NO
53	flight	arrival_airport_name	NO
54	flight	arrival_city	NO
55	flight_norm	flight_number	NO
56	flight_norm	airline_name	NO
57	flight_norm	departure_airport_name	NO
58	flight_norm	departure_city	NO
59	flight_norm	arrival_airport_name	NO
60	flight_norm	arrival_city	NO
61	flights	flight_id	NO
62	flights	sch_departure_time	NO
63	flights	sch_booking_id	NO

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Query

```
1 SELECT table_name, column_name, is_nullable
2 FROM information_schema.columns
3 WHERE table_schema = 'public'
4 ORDER BY table_name, ordinal_position;
```

Query History

Scratch Pad

Data Output Messages Notifications

	table_name	column_name	is_nullable
38	booking	ticket_price	NO
39	booking_flight	booking_flight_id	NO
40	booking_flight	booking_id	NO
41	booking_flight	flight_id	NO
42	booking_flight	created_at	NO
43	booking_flight	updated_at	NO
44	booking_norm	booking_id	NO
45	booking_norm	passenger_full_name	NO
46	booking_norm	passenger_passport_num	NO
47	booking_norm	ticket_number	NO
48	booking_norm	ticket_price	NO
49	flight	flight_number	NO
50	flight	airline_name	NO
51	flight	departure_airport_name	NO
52	flight	departure_city	NO
53	flight	arrival_airport_name	NO
54	flight	arrival_city	NO
55	flight_norm	flight_number	NO
56	flight_norm	airline_name	NO
57	flight_norm	departure_airport_name	NO
58	flight_norm	departure_city	NO
59	flight_norm	arrival_airport_name	NO
60	flight_norm	arrival_city	NO
61	flights	flight_id	NO
62	flights	sch_departure_time	NO
63	flights	sch_arrival_time	NO

Total rows: 91 Query complete 00:00:00.056

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Query

```
1 SELECT * FROM flight_norm;
2 SELECT * FROM booking_norm;
3 SELECT * FROM seat_assignment_norm;
```

Query History

Scratch Pad

Data Output Messages Notifications

	booking_id	seat_number
1	1	12A
2	1	12B
3	2	14C
4	3	5A
5	4	20F
6	5	20E

Total rows: 6 Query complete 00:00:00.053

Successfully run. Total query runtime: 53 msec. 6 rows affected.

Successfully run. Total query runtime: 75 msec. 6 rows affected.